

Chapter 153 — Epidemiology, Aetiology and Prevention of Prostate Cancer

EPIDEMIOLOGY AND AETIOLOGY

- **Long natural history, often decades;** risk factors may act at different stages of carcinogenesis (initiation → promotion → progression), which makes them hard to identify in population studies.
- **The optimal end point is unclear.** In most cancers, diagnosis is the relevant end point; in prostate cancer it may not be.
- **Prostate-specific antigen (PSA) screening is itself one of the strongest risk factors for diagnosis** since the late 1980s/early 1990s; factors that predict access to care and health-seeking behaviour therefore predict "risk".
- **"Incident" cancer is rarely truly incident:** diagnosis designates a cancer present for years. Screening timing and intensity alter the spectrum of cases, the point in the natural history and the age at diagnosis. Of two men with similar latent cancers, only the one who is screened and biopsied is diagnosed.
- Consequence: PSA-era studies may identify **determinants of diagnosis rather than aetiological factors**.
- **Preferred end point: aggressive disease** — advanced stage at diagnosis, or progression to metastasis or death. Risk factors may act only on cancers able to progress, which are the clinically important ones.
- **Lifestyle evidence is almost entirely observational** (diet, physical activity, body weight, alcohol, tobacco) and cannot prove causality. Causality is judged against criteria set, ideally, before analysis: consistency across diverse groups, adjustment for confounders, freedom from bias (especially detection bias), magnitude, biological plausibility.
- Few studies test **change** in lifestyle, so even for causal factors it is unknown at what age risk is reversible; observational work should establish **when in life** a factor operates.

Incidence, Prevalence and Mortality of Prostate Cancer

- **Second commonest non-skin cancer and fifth leading cause of cancer death** in males worldwide.
- 2020 (185 countries): **~1.4 million cases and 375,000 deaths** — 7.3% of all cancer cases, 3.8% of cancer deaths.
- **Global age-standardised incidence rate (ASR) 31/100,000; lifetime cumulative risk 3.9%.**
- **First for incidence in over half of countries (112/185); second cause of cancer death in 48 countries, after lung cancer.**
- By Human Development Index (HDI): **ASR 37.5 in high/very-high HDI** countries (e.g. Switzerland, United States) vs **11.3 in low HDI** (e.g. Niger, Yemen).
- **Incidence varies ~13-fold** (6.3 South Central Asia → 83.4 Northern Europe); **mortality ~9-fold** (3.1 South Central Asia → 27.9 Caribbean).
- Since 2000, incidence is rising in 65 countries, stable in 25 and falling in 45.
- Drivers of variation: **diagnostic practice and PSA use**, reporting quality and absent registries, heritability, environment.

TABLE 153.1 AGE-STANDARDIZED INCIDENCE AND MORTALITY RATES GLOBALLY BASED ON 2020 GLOBOCAN DATA

Incidence band (per 100,000)	Region — incidence / mortality per 100,000
Highest (>70)	Northern Europe 83.4 / 13.0 • Western Europe 77.6 / 9.8 • Caribbean 75.8 / 27.9 • Australia/New Zealand 75.8 / 10.3 • Northern America 73.0 / 8.3

Incidence band (per 100,000)	Region — incidence / mortality per 100,000
Intermediate (40–70)	Southern Africa 65.9 / 20.5 • South America 62.5 / 22.0 • Southern Europe 59.1 / 13.6 • Polynesia 56.5 / 18.8 • Eastern Europe 46.4 / 13.7 • Central America 43.8 / 10.9 • Middle Africa 40.8 / 24.8
Low (<40)	Melanesia 35.8 / 17.0 • Western Africa 33.1 / 20.2 • Western Asia 28.6 / 8.4 • Eastern Africa 27.9 / 16.3 • Eastern Asia 16.8 / 4.7 • Northern Africa 16.6 / 8.2 • South-Eastern Asia 13.5 / 5.4 • South Central Asia 6.3 / 3.1
Overall	31 / 7.7

From GLOBOCAN 2020 (gco.iarc.fr).

- **What it is for:** showing that incidence and mortality are **decoupled**.
- **Pattern:** incidence tracks **screening intensity and HDI** (Northern/Western Europe, Northern America, Australia/New Zealand); mortality peaks where **African ancestry and limited access** coincide (Caribbean, Middle Africa, South America, Southern and Western Africa).
- **Trap:** Northern America is top-five for incidence yet among the **lowest for mortality (8.3)**; the Caribbean is joint third for incidence but **first for mortality**. Asia is lowest on both.

Europe

- **Commonest cancer in Europe and third leading cause of cancer death.**
- Overall ASR 63; Northern 83.4, Western 77.6, Southern 59.1, Eastern 46.4 — variation attributed mainly to **diagnostic practice**.
- Incidence highest in Northern/Western and lowest in Eastern European countries; mortality highest in Northern Europe.
- 2020: ~108,000 deaths; mortality ASR 11.1. Mortality has fallen ~7% since 2015 with therapeutic advances — **Poland the only country with rising mortality**.

North America

- United States, Canada and Mexico combined: ASR incidence 73, mortality 8.3.

United States.

- **Commonest solid tumour in males.** Data principally from the National Cancer Institute's Surveillance, Epidemiology and End Results (SEER) programme, population-based since 1975.
- Incidence **rising ~3%/year (2014–2019)** after two decades of decline.
- 2023: ~288,300 cases and 34,700 deaths; **29% of all male cancer diagnoses**.
- **Lifetime risk ~1 in 8 (12.6%).** Probability by age: 0.2% <50 → 1.8% at 50–59 → 5.2% at 60–69 → 9.2% at ≥70.
- ASR incidence 72, mortality 8.2 (2020).
- **5-year survival 97%, yet second leading cause of cancer death** (11% of cancer deaths).
- Metastatic presentation rising since 2012, especially in males under 70; burden projected to rise 42% by 2025.
- Black males: prostate cancer is **37% of cancer diagnoses and 17% of cancer deaths**. Disparity with White males is multifactorial, including unequal access to care.

Canada.

- Commonest solid tumour in males; **lifetime risk ~1 in 9**; 20% of new male cancers; 5-year survival 91%; ASR incidence 80, mortality 8.9.

- Incidence rose 1969–2010 with **peaks in 1993 (PSA introduced) and 2011 (updated screening recommendations)**, then fell; it kept rising under 70 and fell at ≥ 70 .

Mexico.

- Global Burden of Disease 2019: ASR incidence 52.3, mortality 19.4.

Latin America and the Caribbean

- Heterogeneous rates, mostly from differences in screening and reporting; trend data limited.
- South America: ASR incidence 62.5, mortality 22.0. Incidence rising in Colombia, Ecuador and Costa Rica, stable in Brazil. Mortality rising in Brazil, Colombia and Ecuador; falling in Argentina, Costa Rica and Chile; lowest in El Salvador, Bolivia and Costa Rica.
- **Caribbean: third-highest incidence (75.8) and highest mortality in the world (27.9)**. Highest national incidence: **Martinique (182.0) and Guadeloupe (176.6)**; also Jamaica, Barbados, Tobago. Highest mortality: Barbados, Jamaica, Haiti.
- **Afro-Caribbean males** have more aggressive disease and worse outcomes — socioeconomic constraint, later-stage diagnosis and possibly genetics.

Hispanic Ethnicity

- United States Hispanic/Latino males present with **more advanced disease, higher grade and higher PSA** than non-Hispanic Whites, yet have **lower prostate cancer mortality**.
- Hispanic migrants have lower prostate cancer mortality than both non-Hispanic Whites and United States-born Hispanics.
- **"Hispanic paradox"**: comparable or better outcomes despite disparities in access and socioeconomic status.

Africa

- ~93,000 cases and ~47,000 deaths — a **high death-to-case ratio**.
- ASR: Southern 65.9 > Middle 40.8 > Western 33.1 > Northern 16.6.
- Variation from screening, medical access, registry completeness and lifestyle. **PSA screening uncommon in sub-Saharan Africa** (single-digit percentages, e.g. Ghana, Senegal).
- **Westernisation** of diet and lifestyle, with rising obesity, is a proposed contributor.

Asia

- **Historically lowest incidence**: ASR 14; ~371,000 cases (2020). South Central 6.3 < South-Eastern 13.5 < Eastern 16.8 < Western 28.6.
- Highest national incidence: **Japan, Israel, Armenia**. Incidence rising — fastest in the Republic of Korea (up to 13.8%/year) — from PSA screening and better biopsy technique.
- **Mortality lowest of all regions** (ASR 4.4); highest in Armenia, Georgia, Turkey; rising in the Philippines, Korea, Thailand, China and Iran — obesity, Western diet, inactivity and better reporting.
- **Migrant effect**: risk in native Asians is one-third lower than in Asian Americans — less screening in Asia plus Western diet and lifestyle.

Oceania

- **Australia/New Zealand: ASR 75.8** (level with the Caribbean, among the highest worldwide); mortality 10.3.
- Polynesia: **lower incidence (56.5) but higher mortality (18.8)**.
- Highest incidence in the region: **New Caledonia**.

Fig. 153.1 All-age age-standardised incidence (A) and mortality (B) rates by geographic region

- World choropleth maps from GLOBOCAN 2018 data (McHugh et al., 2022). Incidence bands run from ≥ 69.0 to < 10.6 per 100,000; mortality from ≥ 20.7 to < 4.7 .
- Incidence is darkest across Northern America, Northern/Western Europe and Australia; mortality is darkest across **sub-Saharan Africa and the Caribbean** — the visual form of the incidence–mortality decoupling.

Concept — Incidence measures diagnostic intensity as much as disease, whereas mortality reflects biology and access; screened, high-HDI populations have the highest incidence, and populations of African ancestry with limited access have the highest mortality.

KEY POINTS: EPIDEMIOLOGY AND AETIOLOGY

- Long natural history (decades) makes epidemiological study difficult.
- Lifestyle evidence is observational and cannot prove causality.
- Incidence and mortality vary substantially by region; first for incidence in 112/185 countries, second cause of cancer death in 48.

PROSTATE-SPECIFIC ANTIGEN SCREENING

- **PSA is made almost exclusively by the prostate** and reaches blood by leakage, so **anything that increases leakage raises PSA**: benign enlargement, inflammation and cancer.
- Mainstay of early detection since the mid-to-late 1980s; still controversial. Whether a man is screened has a **large effect on his measured risk and mortality**.

Impact on Incidence

- **PSA screening is one of the strongest predictors of detection.**
- Best measured in the **European Randomized Study of Screening for Prostate Cancer (ERSPC)**: >182,000 men randomised to PSA every 2–4 years (by country) vs no screening.
- **Excess incidence in the screened arm shrinks with time** as controls are diagnosed later, at a later stage; unrecorded screening in controls may add to the erosion.

ERSPC VS PLCO — CONTROL-ARM CONTAMINATION

	ERSPC	Prostate, Lung, Colorectal and Ovarian (PLCO) Screening Trial
Setting	Multiple European countries	United States
Background screening	Uncommon — few controls PSA-tested	Frequent — control arm heavily screened
Consequence	Clean contrast ; isolates the effect of screening	Contaminated contrast; effect diluted

Impact on Mortality**ERSPC — INCIDENCE AND MORTALITY OVER FOLLOW-UP**

Follow-up (years)	Excess incidence, screened arm	Reduction in prostate cancer death
1–9	85%	16%
1–11	60%	22%
1–13	49%	—

Follow-up (years)	Excess incidence, screened arm	Reduction in prostate cancer death
1–16	36%	~20% (plateau)

- **Pattern:** the incidence excess decays while the mortality benefit **builds and then plateaus** — benefit is time-dependent.
- **Intention-to-screen underestimates benefit:** where ITT showed 20%, correcting for non-attendance gave 27%, and also for contamination 31%.
- **Bottom line: screening vs none reduces prostate cancer death by ~20–30%.** Most deaths are therefore not preventable by PSA alone — better screening, treatment **and primary prevention** are needed.

Impact on Prostate Cancer Epidemiology

- Prevention requires first understanding the factors linked with risk.
- "In a population with widespread PSA screening, a diagnosis of prostate cancer signifies, in most cases, that a man happened to have a PSA test at that time."
- **Healthy-user confounding:** men with healthy habits (diet, exercise, non-smoking, lean) also seek PSA tests. Unadjusted, **healthy behaviours appear to raise incidence yet lower mortality**. This likely confounds much of the risk-factor literature of recent decades, forcing reliance on the few randomised trials.

Impact on Prevalence

- Incidence and prevalence estimates are biased by changing screening practice; **autopsy studies of latent cancer are less biased**.
- **Latent prevalence rises steeply with age:** ~5% <30 years → 15% at 40–50 → 22% at 50–59 → 29% at 60–69 → 59% >79.
- **Race:** at 70–79, latent prevalence ~36% in White vs ~51% in Black males.
- Stable over time globally, but **rising in Asia, notably Japan**.
- Limitations: study quality, few autopsies, poor standardisation — more high-quality, diverse studies needed.

Concept — *PSA screening inflates and advances incidence and confounds every lifestyle association; mortality, aggressive disease and autopsy prevalence are the end points that escape its bias.*

KEY POINTS: PROSTATE-SPECIFIC ANTIGEN SCREENING

- Benign enlargement, inflammation and cancer all raise PSA.
- PSA screening is one of the strongest predictors of detection and reduces prostate cancer death by ~20–30%.
- Unadjusted, healthy behaviours appear to raise incidence but lower mortality.

RISK FACTORS

- Multifactorial cause. **Established risk factors: advanced age, race and family history.**

Age

- **~60% diagnosed at ≥65; mean age at diagnosis 66.** Risk low <40, rising steadily thereafter.
- Historical (1974–1994) probability of diagnosis: 0.005% at ≤39, 2.2% at 40–59, 13.7% at 60–79.

- **Rising incidence in adolescents and young adults** (aged 15–39): ~2%/year from 1990 to 2017, across all continents.
- **55% of prostate cancer deaths occur at ≥65.** Over 20 years of follow-up, the risk of dying of prostate cancer is higher when diagnosed at ≥70 than below 70.
- An ageing population → growing burden and survivorship needs.

Race

- **Highest incidence in Black males** (172.6/100,000 in the United States).
- **Prostate cancer-specific mortality twice that of White males** — more metastatic presentation and younger age at diagnosis.
- Drivers: **structural racism, regional segregation, unequal access**; these also raise comorbidity (cardiovascular disease, diabetes) linked with poor cancer outcomes.
- Genetics: **8q24 risk alleles**, identified in genome-wide association studies (GWAS), are more common in African ancestry. Genomic testing can identify aggressive subsets for targeted intervention.
- **Equal access → equal outcome**: with similar access and treatment, Black patients with non-metastatic disease have prostate cancer-specific mortality similar to White patients.
- Autopsy studies show higher latent cancer rates from very early ages.

Family History

- ~20% of patients have a family history.
- **First-degree relative (father or brother) → 2- to 3-fold risk**, rising with the number of affected relatives and with **younger age at their diagnosis**.

FAMILY HISTORY — RISK GRADIENT

Family history	Relative risk vs none
Any first-degree relative	2–3-fold
Two affected first-degree relatives	~3.5-fold
Both father and brother	9-fold
Relative diagnosed <60 vs ≥60	~2.5-fold vs ~1.6-fold
Two or more relatives diagnosed <60	~5.7-fold
Father or brother died of prostate cancer	2-fold risk of prostate cancer death

- **What it is for**: risk scales with **number** of relatives, **closeness** and **youth** at diagnosis; the **lethality** of the relative's cancer is itself heritable.

Shared Lifestyle

- Familial clustering may partly reflect **shared diet, activity, screening behaviour, smoking, alcohol and environmental toxins**.
- Family history drives PSA testing in relatives, but a Swedish study concluded familial aggregation is **not substantially due to shared screening**.
- Shared modifiable factors are targets for family and community interventions.

Genetics

- **Genetics outweigh shared lifestyle: heritability up to 58% (twin studies)**.
- **BRCA1/BRCA2** (hereditary breast-ovarian cancer genes) → aggressive prostate cancer.
- **HOXB13 (homeobox B13)** — prostate-specific; hereditary prostate cancer.

- **PCGEM1** — prostate-specific; overexpressed in most tumours. **DD3** — marker for early detection.

Homologous Recombination DNA Repair

- **Germline DNA-repair mutations (BRCA1, BRCA2, ATM, CHEK2, PALB2) → higher risk and more aggressive cancer.**
- Clinical uses: targeted early screening (BRCA1/2, ATM, mismatch repair genes, HOXB13, CHEK2); caution in active surveillance for BRCA1/2 and ATM carriers; treatment selection in advanced disease — androgen receptor (AR)-targeted agents, poly(ADP-ribose) polymerase (PARP) inhibitors (BRCA1/2, ATM), immune checkpoint inhibitors (mismatch repair).

Mismatch Repair Genes

- Mismatch repair (MMR) mutations maintain genome stability; **uncommon in prostate cancer (2–5%)**, so evidence is limited.
- **MLH1, MSH2, MSH6, PMS2 (Lynch syndrome)** are associated with prostate cancer.
- MMR-deficient metastatic disease shows **high sensitivity to androgen deprivation therapy (ADT)**.

Single-Nucleotide Polymorphisms

- GWAS have found **>200 common single-nucleotide polymorphisms (SNPs)**, each of low-to-moderate effect.
- Strongly linked with progression: **HSD3B1** (hydroxy-delta-5-steroid dehydrogenase; also susceptibility), **CYP19A1, HSD17B4**. SNPs in **KLK3 (the PSA gene)** are linked with risk.

Genomic Regions

- Key loci **8q24 and 17q21**, important for risk differences across ancestry.
- **8q24**: more common in Black individuals; also associated in European populations. **17q21**: present in 4–7% of those of African descent vs <1% of non-African populations.
- Asia: **ERG oncoprotein-positive cancer is uncommon in East Asian males; FOXA1 mutated more often in Chinese males.**
- Multi-ancestry GWAS meta-analysis: 86 new variants; risk scores differ across European, African and Asian ancestry.

GERMLINE GENES AND LOCI — ASSOCIATION AND CLINICAL IMPLICATION

Gene / locus	Class	Association	Clinical implication
BRCA2, BRCA1	Homologous recombination	Higher incidence; BRCA2 familial; aggressive disease	Targeted screening; PARP inhibitor; caution in surveillance
ATM	Homologous recombination	Metastatic disease	Targeted screening; PARP inhibitor; caution in surveillance
PALB2	Homologous recombination	Familial; aggressive and lethal disease	—
CHEK2	Regulates TP53 and DNA repair	Familial disease	Targeted screening
MLH1, MSH2, MSH6, PMS2	Mismatch repair (Lynch)	Prostate cancer (2–5% of cases)	Immune checkpoint inhibitor; ADT-sensitive
HOXB13	Prostate-specific homeobox	Hereditary disease	Targeted screening
8q24 / 17q21	GWAS loci	More frequent in African ancestry	Explain part of the racial excess

Gene / locus	Class	Association	Clinical implication
HSD3B1, CYP19A1, HSD17B4	Steroid-pathway SNPs	Progression	—

Polygenic Risk Scores

- Polygenic risk scores (PRS) combine risk alleles into a personalised risk to guide **screening decisions**.
- **They predict whether prostate cancer develops, not whether it is aggressive.**
- A multi-ethnic PRS stratifies risk in European and African ancestry. PRS for aggressive disease and survival, and trials of clinical utility, are needed.

Concept — Age, ancestry and family history are the only unequivocal risk factors, and genetics carries most familial risk; the germline repair genes that matter clinically predict **aggressive** cancer, which is why they change screening and treatment while polygenic scores, which predict only **any** cancer, do not yet.

KEY POINTS: RISK FACTORS

- Established risk factors: advanced age, race and family history.
- Germline homologous recombination mutations (BRCA1, BRCA2, ATM, CHEK2, PALB2) raise risk and aggressiveness.
- MMR mutations (MLH1, MSH2, MSH6, PMS2; Lynch syndrome) are associated with prostate cancer.

POSSIBLE RISK FACTORS

- Body composition (height, obesity, waist circumference) and comorbidities; findings often contradictory. Diet and activity are covered separately.

Height

- **Taller height → higher risk**, supported by recent studies and meta-analyses; ~5% higher risk per 10-cm increment, with estimates varying by measurement and population.
- Association is **with high-grade disease and prostate cancer death, not low-grade disease** (including in the European Prospective Investigation into Cancer and Nutrition, EPIC).
- Height reflects childhood/adolescent hormones and diet; **taller height predicts progression in Black but not White males**.

Obesity

OBESITY — TOTAL VS AGGRESSIVE PROSTATE CANCER

Total incidence	Aggressive disease
Mixed — positive overall in meta-analyses, but individual studies show strong, null or even protective associations	Consistently increased
Body mass index (BMI) inversely associated with non-metastatic low-grade cancer	BMI positively associated with high-grade and metastatic/fatal cancer
PCPT and REDUCE: obesity → lower low-grade risk	PCPT and REDUCE: obesity → higher high-grade risk

- Dissent: one study found BMI ≥ 30 kg/m² lowered risk regardless of grade, while **waist circumference raised it**.
- **Life course:** weight gain from early adulthood to midlife is mostly not associated, though a multiethnic cohort found adiposity change did raise risk.

- **Around diagnosis:** weight loss is protective; **weight gain raises recurrence after prostatectomy.**
- **Waist circumference** (abdominal obesity): positive for advanced disease in a few studies; null in the Health Professionals Follow-up Study (HPFS); inverse for total and localised disease and null for advanced disease and mortality in Swedish cohorts.
- Mechanisms: **insulin/insulin-like growth factor 1 (IGF-1), sex hormones, adipokine signalling** (Fig. 153.2).
- ~650 million obese adults worldwide and rising — the link with aggressive disease is a **public-health concern.**

Comorbidities Possibly Associated With Prostate Cancer

- Metabolic syndrome, hyperlipidaemia, hypertension and cardiac disease are associated with **higher risk; diabetes with lower.** Older patients commonly carry several, compounding diagnosis and outcome.

Metabolic Syndrome

- Insulin-resistance syndrome; cluster raising cardiovascular and diabetes risk; **prevalence ~12–31% of adults.**
- Mechanisms: altered insulin and IGF-1; sex-steroid shift — **↑ oestradiol, ↓ sex hormone-binding globulin (SHBG), ↓ free testosterone** (Fig. 153.2).
- Meta-analyses (heterogeneous): slight association with incidence and with high-grade disease; significant in **cross-sectional** studies but not in pooled cohort and case-control studies.
- Linked with worse features: **higher Gleason score, positive surgical margins, biochemical failure.**
- Geography: positive in European studies, not in United States or Asian studies.

Hypertension

- Affects ~one-third of adults; leading modifiable cardiovascular risk factor; linked with inflammation.
- **Increased prostate cancer risk;** in a large cohort, men aged ≥45 with **systolic ≥150 mmHg vs <130 mmHg** had higher incidence and, once diagnosed, higher mortality.
- Meta-analyses significant but heterogeneous.

Cardiovascular Disease

- **Leading cause of death, even in prostate cancer patients.** Shared biology (inflammation, atherosclerosis, oxidative stress, hormones, cholesterol) and shared risk factors (age, race, diet, activity, alcohol). Patients with cardiovascular disease are more often diagnosed.
- **ADT and cardiovascular disease:**

ADT AND CARDIOVASCULAR RISK — TRIAL VS OBSERVATIONAL EVIDENCE

Meta-analysis of randomised trials	Observational studies and other meta-analyses
No increase in cardiovascular death; lower prostate cancer-specific and all-cause mortality	Positive association between ADT and cardiovascular disease

- Close monitoring of trial patients may have negated harm → **ADT cardiotoxicity may be preventable.** Baseline cardiovascular assessment and risk-factor management are recommended, especially on ADT.

Diabetes Mellitus

- Raises risk of many cancers (pancreas, kidney, colon) but is **associated with lower prostate cancer risk** (modest reduction in large meta-analyses).
- Mechanisms: **↓ testosterone, ↓ IGF-1, lower PSA (underdiagnosis), metformin use.**

COMORBIDITIES — DIRECTION AND MECHANISM

Condition	Prostate cancer risk	Proposed mechanism / caveat
Metabolic syndrome	↑ (slight); worse pathology	Insulin/IGF-1; ↑ oestradiol, ↓ SHBG, ↓ free testosterone; heterogeneous by region and design
Hypertension	↑	Inflammation; heterogeneous
Cardiovascular disease	↑ diagnosis	Shared biology and risk factors; ADT adds cardiovascular risk
Diabetes mellitus	↓	↓ testosterone, ↓ IGF-1, ↓ PSA, metformin

Fig. 153.2 Proposed framework linking diet, physical activity and body weight to hormones, inflammation and increased prostate cancer risk

- Modified from Giovannucci, 2018. Sequence: **poor diet + physical inactivity → excess adiposity → ↑ leptin/↓ adiponectin, ↑ insulin, ↑ inflammation, ↑ oestrogen → ↓ SHBG, ↑ IGF bioactivity, ↑ bioavailable oestrogen → ↑ proliferation, ↓ apoptosis → ↑ prostate cancer risk.**

Concept — *Body size and metabolic state act on progression more than initiation: height and obesity track aggressive disease, whereas diabetes lowers androgens, IGF-1 and PSA — hence its paradoxically lower risk.*

KEY POINTS: POSSIBLE RISK FACTORS

- Taller height is associated with higher risk.
- Obesity: mixed for diagnosis, consistent for aggressive disease.
- Metabolic syndrome, hyperlipidaemia, hypertension and cardiac disease raise risk; diabetes lowers it.
- Observational data link ADT with cardiovascular disease.

DIET

1. **Population evidence:** international variation, secular trends within populations and rising rates in **migrants** from low- to high-incidence countries imply external (non-genetic) factors. Per capita fat intake correlates with national rates — hypothesis-generating only, as inactivity and PSA screening co-vary with diet.
 2. **Biological plausibility:** food carcinogens formed in processing/cooking damage DNA; some compounds induce enzymes that activate or deactivate carcinogens; oxidative DNA damage is increased by polyunsaturated fat and excess iron and opposed by antioxidants; B vitamins support DNA synthesis, repair and methylation; non-mutagenic routes act through **insulin and IGFs**.
- **Methodological challenges:** measuring intake over decades; many levels of analysis (patterns, macronutrients, foods, micronutrients, phytochemicals); foods act in combination; correlated intakes (meat with potatoes; fortified milk supplies vitamin D, calcium and phosphorus together); **fat-soluble compounds (vitamin D, lycopene) need fat co-ingestion** — vitamin D in fortified skim milk is poorly absorbed compared with whole milk.
 - **Randomised trials** are most rigorous but rare, test single micronutrients and have been **mostly null** — the **Selenium and Vitamin E Cancer Prevention Trial (SELECT)** (selenium 200 µg, vitamin E 400 IU) showed no benefit. A single high-dose micronutrient is unlikely to show a population effect; whole-diet change is more plausible but hard to sustain.
 - **Prospective cohorts** (tens of thousands followed for years) are feasible but confounded; causality is judged by consistency, dose-response, confounder control and plausibility.

- **World Cancer Research Fund/American Institute for Cancer Research (WCRF/AICR)**: expert-graded reviews each decade since 1997 (convincing → probable → limited/suggestive → unlikely). **2018: no dietary factor reached strong evidence for prostate cancer** (literature to 2015); emphasis has since shifted to **dietary patterns**.

KEY POINTS: DIET

- International variation implies non-genetic factors; dietary factors could act at many steps of pathogenesis.
- Causal inference is difficult, and large randomised nutritional trials have generally been null.

Dietary Patterns

THREE STRATEGIES FOR DEFINING DIETARY PATTERNS

Strategy	Basis	Examples
A priori	Guidelines, regional diet or prevailing hypothesis	Healthy Eating Index; Mediterranean diet
A posteriori	Data-driven — principal components or cluster analysis	"Western" (unhealthy) vs "Prudent" (healthy)
Biomarker-based	Foods predicting a specified biomarker	Hyperinsulinaemic or inflammatory diet scores

- **A priori**: Healthy Eating Index–2005, Alternative Healthy Eating Index–2010 and alternative Mediterranean score — no association with total, advanced or fatal cancer; Mediterranean meta-analysis null. **No strong support**.
- **A posteriori**: no association for healthy or unhealthy patterns; too few studies of vegetarian variants.
- **Plant-based**: vegan diet associated with lower risk in a large Adventist cohort; reviews inconsistent (definitions vary). In HPFS, a higher plant-based index → **lower fatal prostate cancer**, driven more by **avoiding animal foods** than by plant foods themselves.
- Early evidence: diets lowering **insulin and inflammation** may lower risk.

Macronutrients

- **Fat**: pre-PSA-era studies (Seventh-day Adventists; HPFS) linked animal fat with risk; **PSA-era studies (EPIC; meta-analysis of cohorts) are null** for total, saturated and unsaturated fat and for total or advanced disease. WCRF/AICR 2018: fat, protein and carbohydrate "limited/no conclusion".
- **Carbohydrate**: **sugar-sweetened beverages → increased risk**; glycaemic index/load null.
- **Dairy**: the non-fat milk fraction (protein, minerals) **raises IGF-1**, a risk factor.

WCRF/AICR GRADING — THE PRE-PSA VS PSA-ERA DOWNGRADES

Exposure	2007 report	2018 report
Dairy products	Probable (↑ risk)	Limited/suggestive
Calcium	Possible/probable (↑ risk)	Limited/suggestive
Tomatoes/lycopene	Probably causal (↓ risk)	No conclusion possible
Fat, protein, carbohydrate	—	Limited/no conclusion

- **What it tests**: downgrading "strong" evidence is historically unusual; **three downgrades in one period** suggest that cancers diagnosed in the pre-PSA and PSA eras differ, altering risk-factor relationships.

Carcinogens From Well-Done Meat

- **Heterocyclic amines** form from creatine/creatinine, sugars and amino acids when meat or fish is cooked **above 130 °C** for long periods — grilling, barbecuing, frying, broiling.
- Questionnaire estimates are unreliable. Cohorts: **null for total** cancer, but most suggested an association with **aggressive/advanced** disease → biomarkers needed.

Fatty Fish and ω -3 Fatty Acids

- ω -3 polyunsaturated fatty acids are anti-inflammatory. Dietary docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA) — null for total cancer; circulating levels unclear.

CIRCULATING Ω -3 — PSA ERA VS PRE-PSA ERA

SELECT (bloods 2001–2004, PSA era)	Physicians' Health Study (bloods 1982, pre-PSA era)
Higher plasma long-chain ω -3 → increased total and high-grade cancer; mostly localised, screen-detected	Higher serum long-chain ω -3 → protective

- Fish/seafood: **no association with total, inverse with fatal** prostate cancer (confirmed by meta-analysis).
- A 6-year ω -3 randomised trial: no effect on incidence; underpowered for fatal disease.

Tomato Products and Lycopene

- Among fruits and vegetables, **only tomato products** were associated with lower risk; focus fell on **lycopene** (red carotenoid; also pink grapefruit, watermelon).
- Effect confined to **processed/cooked tomato, especially sauce**: lycopene in raw tomato is tightly bound and lipophilic — **bioavailable only when cooked in oil**.
- 2003 meta-analysis: high tomato intake and high serum lycopene each associated with modestly lower risk. WCRF/AICR: "probably causal" (2007) → "no conclusion possible" (2018); the earlier studies were pre-PSA.
- **HPFS**: lycopene → lower **lethal (metastatic)** cancer in both eras, but lower **total** cancer **only pre-PSA** — pre-PSA cancers were mostly aggressive, PSA-era cancers mostly indolent. Lycopene intake → **lower angiogenesis** in prostatectomy tumours; angiogenesis strongly predicts progression.
- Pooled analysis of 15 studies: serum lycopene **inverse with advanced** cancer. Evidence for lycopene **supplements** is limited.

Coffee Consumption

- Antioxidant and anti-inflammatory compounds. Meta-analysis of cohorts: highest intake → modestly lower risk, **stronger for advanced and fatal disease**, in the highest-quality studies and with fuller adjustment (smoking, BMI, PSA screening).
- **HPFS**: inverse association with advanced disease for **both regular and decaffeinated** coffee → **non-caffeine components** responsible.

Alcohol

- Raises risk of at least seven cancers; **acetaldehyde** is an established carcinogen. WCRF/AICR 2018: "limited–no conclusion" for prostate cancer.
- Case-control plus cohort meta-analysis: **dose-response with prostate cancer mortality**, raised at **≥45 g/day ethanol (>3–4 standard drinks)**; a cohort-only meta-analysis found no consistent association.
- **Beverage type**: liquor raises risk at lower doses than wine, which does so only at high doses; red-wine polyphenols may offset ethanol at low intake.

- **Binge drinking** raised total prostate cancer risk where overall intake did not.

Concept — *The PSA era broke the diet literature: associations built on clinically detected, largely aggressive pre-PSA cancers (fat, dairy, lycopene, ω -3) weakened or reversed once indolent screen-detected disease dominated, so dietary signals now appear mainly against **advanced and fatal** disease.*

KEY POINTS: DIETARY PATTERNS AND MACRONUTRIENTS

- A priori dietary-pattern studies do not strongly support a role in prostate cancer risk.
- Dietary fat is not associated with total or advanced prostate cancer.
- WCRF/AICR 2018 downgraded dairy and calcium from "probable" to "limited/suggestive", and tomatoes/lycopene from "probably causal" to "no conclusion possible".

TOBACCO

SMOKING — INCIDENCE VS LETHALITY

Incidence	Fatal / advanced disease
Not increased — often lower in smokers	Increased — acknowledged by the United States Surgeon General; ~40% higher prostate cancer mortality (meta-analysis)
Likely from less PSA screening and biopsy	More advanced at diagnosis — seen even pre-PSA , so not only screening

- Smokers may develop a **more aggressive phenotype**, prone to metastasis and death.
- Mechanisms: tobacco-carcinogen mutations in progression genes; **↑ androgens**; angiogenesis; impaired immunity. Non-causal contributor: delayed diagnosis.
- **Former smokers who quit ≥10 years ago carry no excess aggressive risk** → quitting at any adult age is advisable.

Concept — *Smoking is the model of a factor that **lowers apparent incidence while raising lethality**; only an aggressive-disease end point reveals it.*

KEY POINTS: TOBACCO

- Tobacco does not increase incidence, but smokers are more likely to develop fatal prostate cancer.

PHYSICAL ACTIVITY AND SEDENTARY BEHAVIOUR

- A probable protective factor for several cancers; **unclear for prostate**. Plausible via insulin signalling, inflammation, immunity and weight control. WCRF/AICR: no clear conclusion for incidence.
- Limitations: measurement error; differing effects of light, moderate and vigorous activity and of leisure vs occupational activity; differing effects by subtype (advanced vs organ-confined).
- 2018 meta-analysis: **no clear association with total** cancer; regular recreational activity → **lower advanced/aggressive** cancer (only two studies).
- **HPFS: vigorous (not moderate) activity** → lower **advanced (not total)** cancer, after accounting for PSA screening; repeated 2-yearly measurements improved accuracy.
- Swedish cohort (minimal PSA screening): lifetime occupational activity and walking/cycling → lower total cancer.

- **Sedentary behaviour** may raise aggressive cancer risk, **mediated by adiposity**.

Concept — *Consistent, long-term, vigorous activity is linked with fewer **clinically significant** cancers; as with diet and smoking, the signal appears against advanced disease, not total incidence.*

COMPOSITE LIFESTYLE

- **A composite approach outperforms any single factor.**
- HPFS and Physicians' Health Study (PHS): a healthy-lifestyle score (smoking, activity, body weight, diet) → **30–50% lower lethal prostate cancer**.
- HPFS: adherence to smoking, activity, weight and alcohol guidelines could prevent **21% of fatal cancers** in the cohort and **40%** in United States White patients.
- Highest polygenic-risk quartile: lifetime lethal risk **5.3% with unhealthy vs 1.6% with healthy lifestyle** → **genetic predisposition is not deterministic**.

Concept — *Genes set the baseline, but lifestyle modifies **lethality** even in the highest genetic stratum.*

KEY POINTS: COMPOSITE LIFESTYLE

- A composite lifestyle approach is more effective than any single factor in reducing incidence or mortality.

SUPPLEMENTS, VITAMINS AND MINERALS

- Use is high: over half of United States adults (and about a third of children) in the past 30 days; lower but substantial elsewhere (e.g. South Korea).
- **No strong evidence of benefit for healthy people eating a balanced diet.** United States Preventive Services Task Force (USPSTF): little or no benefit for cancer, cardiovascular disease or death, **except a small benefit in cancer incidence with multivitamins**.
- For prostate cancer: **no consistent benefit, and some harm**. Where large trials in prostate cancer patients exist (secondary prevention), they are noted.

Vitamin D

- Prostate cancer cells and normal prostate express the **vitamin D receptor**; vitamin D induces apoptosis in vitro.
- Serum 25-hydroxyvitamin D meta-analyses conflict; one study found a **U-shaped** relationship (high and low levels both linked with risk).
- No large prostate-prevention trial; pooled trials show no effect on overall cancer. **Vitamin D and Omega-3 Trial (VITAL)**: >25,000 randomised to **2000 IU vitamin D3 daily vs placebo for 5 years** — no significant effect on prostate cancer (192 vs 219 cases, non-significant); **significantly fewer metastatic or fatal cancers overall**, which merits confirmation.
- In prostate cancer patients, six randomised trials: **no effect on PSA or mortality**.

Selenium

- Selenised-yeast trial: primary end point (skin cancer) null, but a large **secondary reduction in prostate cancer** → prompted SELECT.
- **SELECT**: 35,533 men; selenium, vitamin E, both, or placebo; planned 7–12 years; **stopped early (median 5.46 years) for futility**. Selenium alone and with vitamin E: no benefit (hazard ratios >1).
- Counter-arguments: wrong form of selenium; earlier protective signals only in smokers.

Vitamin E

- Rationale from the **Alpha-Tocopherol, Beta-Carotene Cancer Prevention (ATBC) study**: α -tocopherol lowered prostate cancer by 32% (secondary analysis) and prostate cancer death by 41%.
- SELECT: no benefit; **vitamin E alone suggested increased risk ($P = 0.06$)**.

ATBC VS SELECT — WHY VITAMIN E RESULTS DIVERGED

	ATBC	SELECT
α -Tocopherol dose	Tenfold lower	400 IU
Population	Heavy smokers	Almost all non-smokers
PSA screening	Minimal	Widespread
Mortality/incidence ratio	25% (aggressive, clinically detected)	0.5% (indolent, screen-detected)
Result	↓ incidence 32%; ↓ fatal 41%	No benefit; possible harm

- **Verdict**: not recommended for average-risk men; untested whether **low-dose vitamin E lowers mortality in heavy smokers**.

Vitamin C

- Very limited data: an early meta-analysis found dietary vitamin C inversely associated; later studies refute this.
- At high doses vitamin C may act as a **pro-oxidant**; **supplemental antioxidants above the median → increased risk**, including advanced and high-grade disease.

Vitamin K

- Reduces tumour growth preclinically; human observational studies show **no association**.

Calcium

- Bone component and intracellular signal. Often co-prescribed with vitamin D for **ADT-induced osteoporosis** — so its link with risk matters clinically.
- Meta-analyses: dietary and supplemental calcium → increased total (possibly advanced) cancer; another found **dairy (not non-dairy or supplemental calcium)** linked with incidence, but **supplemental calcium linked with fatal disease**. **Phosphorus** may be the dairy component linked with fatal cancer. Serum calcium: mixed.
- **Verdict**: no evidence of benefit; **use calcium supplements cautiously** in those at risk.

Zinc

- Prostate zinc levels are very high relative to the rest of the body; most "prostate health" supplements are high in zinc.
- Meta-analysis: no significant relationship, with a suggestion of harm at higher intake. **HPFS: supplemental zinc ≥ 75 mg/day or for ≥ 15 years → higher lethal and high-grade cancer**.
- **Verdict**: avoid prolonged high-dose zinc.

Phytochemicals

- Naturally occurring plant molecules; plausible given plant-rich diets. **None has shown benefit in a large randomised prevention trial**; several tested in patients with mixed results.

Phytoestrogens

- Oestrogen-like phytochemicals; of interest because prostate cancer is androgen-dependent.

Soy.

- Whole food (protein, milk, fibre); can be **pro- or anti-oestrogenic**. Low Asian risk suggested soy, but many lifestyle factors differ.
- Meta-analyses: soy, genistein, daidzein and fermented soy → lower risk; an update confirmed **soy food (not isoflavones)**.
- Enthusiasm dampened by a **randomised trial: soy protein did not prevent recurrence** in high-risk disease.

Flaxseed

- Rich in fibre, lignans and ω -3; reduced prostate cancer in engineered mice. No risk data; a pre-prostatectomy trial showed **reduced tumour Ki-67** (proliferation marker), without follow-up studies.

Curcumin

- Principal component of turmeric (ginger family); potent in vitro but **poorly absorbed and rapidly metabolised**. Large literature, much of it recent; no population data. Most trials in patients: **no PSA effect**; the positive ones combined curcumin with other herbs.

Ellagitannins

- Found in pomegranate; the chapter groups green tea with them. Ellagitannins are converted by the **gut microbiome into urolithins**, which are absorbed and purportedly anticancer. Studied mainly in patients, not primary prevention.
- **Pomegranate**: single-arm study slowed PSA rise in biochemical recurrence → **placebo-controlled trial failed to confirm**; no tumour effect before prostatectomy (higher urolithin A → less oxidative stress); on active surveillance no effect on PSA, IGF-1 or PSA doubling time, but lower oxidative stress and AR expression.
- **Green tea**: in high-grade prostatic intraepithelial neoplasia (HGPIN) or atypical small acinar proliferation (ASAP), trial meta-analysis showed a **large reduction in progression to cancer**; observationally, a linear dose-response significant only **beyond 7 cups/day** — impractical; extracts need study.

Lycopene

- Systematic review: over half of lycopene-cancer studies concern the prostate. **Dietary studies mostly positive (11 of 20); interventional studies mostly null (4 of 5)**.
- In animals, **tomato powder, but not lycopene**, inhibited cancer → whole-food benefit may not reduce to one agent.
- Fat-soluble: an **ultra-low-fat diet**, even plant-based, may deliver little lycopene — **food combinations matter**.

Resveratrol

- Stilbenoid polyphenol (peanuts, blueberries, cranberries, grape skin, red wine). Mixed preclinical data, including harm; **poor solubility and bioavailability**; 4 months had no effect on PSA in healthy men.

SUPPLEMENTS, VITAMINS AND MINERALS — EVIDENCE AND VERDICT

Agent	Best evidence	Verdict
Vitamin D	VITAL — no prostate effect; ↓ metastatic/fatal cancer overall	Not recommended; signal to confirm
Selenium	SELECT — no benefit, stopped early	No benefit
Vitamin E	ATBC ↓ (smokers, low dose); SELECT — possible harm	Not recommended
Vitamin C / antioxidants	Null; high supplemental antioxidants ↑ risk	Avoid high-dose
Vitamin K	Observational null	No benefit

Agent	Best evidence	Verdict
Calcium	↑ total; supplements ↑ fatal	Use cautiously
Zinc	High-dose/prolonged ↑ lethal and high-grade	Avoid prolonged high dose
Soy	Observational ↓; randomised trial null for recurrence	Unproven
Flaxseed	↓ Ki-67 before prostatectomy	Unproven
Curcumin	No PSA effect alone	Unproven
Pomegranate	Placebo-controlled trial null	Unproven
Green tea	↓ progression from HGPIN/ASAP; >7 cups/day	Promising; extracts to test
Lycopene	Diet positive; interventions null	Whole food over supplement
Resveratrol	No PSA effect	Unproven

Concept — *Single agents extracted from a protective diet and given in high doses have consistently failed or caused harm (SELECT, calcium, zinc, antioxidants); any benefit belongs to **whole foods and their combinations**.*

KEY POINTS: SUPPLEMENTS, VITAMINS AND MINERALS

- No strong evidence that any supplement, vitamin or mineral benefits healthy people eating a balanced diet.

ASSORTED RISK FACTORS

Transgender Identity

- Transgender females (assigned male at birth) **retain the prostate after gender-affirming surgery** and remain at risk.
- Claims-based and chart-review studies: **incidence significantly lower** than in cisgender males.
- Those on **gender-affirming hormone therapy** show the most aggressive markers (higher PSA density, more high-grade disease) — delayed diagnosis or selection of **androgen-deprivation-resistant**, more aggressive clones.
- Reasons: unawareness that a prostate is present, lower screening, oestrogen suppression of cancer development, and **false reassurance from "normal" PSA** despite castrate testosterone.
- **Median PSA 0.02 ng/mL on hormones vs 1 ng/mL in cisgender males (fiftyfold lower) → PSA cut-points must be adjusted.**

KEY POINTS: TRANSGENDER IDENTITY

- Incidence appears significantly lower in transgender females than in cisgender males.

Vasectomy

- **Detection bias:** urologists who perform vasectomy also screen for prostate cancer.
- Meta-analysis: modest overall excess; after **PSA adjustment**, only a very small excess of localised cancer and **no link with high-grade, advanced or lethal disease**.
- **American Urological Association (AUA):** clinicians need not routinely discuss prostate cancer in pre-vasectomy counselling — **vasectomy is not a risk factor**.

Sunlight

- Sunlight converts 7-dehydrocholesterol → previtamin D, activated in liver and kidney (possibly also the prostate).
- **Darker skin absorbs less ultraviolet radiation → lower vitamin D** at equal exposure — a proposed contributor to Black-White disparities.
- Meta-analysis: highest vs lowest sun exposure → significantly lower risk (moderate exposure non-significant); no aggressiveness data; weigh against **skin cancer**.

Ejaculation Frequency

- Higher frequency → **lower risk** (proposed clearance of infectious insults, less inflammation); "lower-risk" thresholds range from $\geq 4/\text{month}$ to $\geq 21/\text{month}$.
- Gene expression in **benign prostate, not tumour**, varies with frequency.
- Alternative: conditions limiting ejaculation may be causal — frequency then an **indirect marker**.

Sexually Transmitted Infections

- Most sexually transmitted infections (gonorrhoea, chlamydia) classically do not affect the prostate but may **spike PSA**. Not a major clinical factor; a meta-analysis suggests STIs overall, and **gonorrhoea** specifically, may raise risk.

Human Papillomavirus

- Human papillomavirus (HPV) is linked with cervical, head and neck, anal and penile cancer.
- Meta-analysis: **modest (~28%) increase**, with heterogeneity and publication bias; contrast **~95% of cervical cancer attributable to HPV**. Given prostate cancer's frequency, even a modest real effect would matter.

Chronic Inflammation

- Inflammatory infiltrates are **near-universal**: in a prevention-trial biopsy series most men had chronic inflammation, with or without cancer.
- **Inflammation raises PSA** → spurious detection-driven link.
- Genetic, molecular and pathological evidence nonetheless supports a **causal link** → rationale for **anti-inflammatory chemoprevention**.

Chronic Prostatitis

- Clinical pain syndrome affecting urinary and sexual function; **episodic, usually without an identifiable organism**. Meta-analysis shows increased risk, but **PSA detection bias** leaves the implication unclear.

Sleep

- Bidirectional relationship with health. **Insomnia → modest increase in risk** (>200,000 people); metabolic disturbance is a plausible mediator — recommend good sleep regardless.
- More sleep disturbance among Black males (shift work, discrimination, stress, noisy neighbourhoods) may contribute to their higher risk.

Shift Work

- **International Agency for Research on Cancer (IARC)/World Health Organization (WHO) 2007: circadian-disrupting shift work "probably carcinogenic to humans"** — based mainly on breast cancer.
- Circadian disruption enhances carcinogenesis in animals; night work inverts the clock, but daylight still entrains it → incomplete sleep — a model for sleep effects.
- Some studies link shift work with **aggressive** disease; meta-analysis evidence weak.

Environment Exposure

ENVIRONMENTAL AND OCCUPATIONAL EXPOSURES

Exposure	Source / use	Association
Per- and polyfluoroalkyl substances (PFAS)	Grease-, oil-, water- and heat-resistant " forever chemicals "	No strong link ; positives often non-linear
Cadmium	Batteries, pigments, metal coatings, plastics; occupational	No link
Organochlorine pesticides	Agriculture and mosquito control 1940s–1960s; DDT ; lindane (lice)	Modest ↑ ; stronger in prospective cohorts
Chromium	Stainless steel, chromium plating	Modest ↑ (largest of the four)

Timing of Puberty

- Cancer found at autopsy in **20-year-olds** implies early-life determinants.
- Puberty onset is poorly defined in males and poorly recalled → mixed literature; **Mendelian randomisation: early puberty → higher risk, especially aggressive.**
- **Earlier puberty in young Black males** is proposed to contribute to more aggressive disease.

Hormonal Levels

- Hormones act at distant and local sites; **advanced cancers synthesise their own testosterone** in a castrate environment.

HORMONES AND PROSTATE CANCER RISK

Hormone	Evidence	Direction
Testosterone	Adult serum (total, free): minimal or protective. Mendelian randomisation: free (not total) testosterone ↑ aggressive and total cancer — positive <60, inverse at older ages. Vertex baldness (not other patterns) → higher risk	↑ (modest, genetic/functional markers)
Insulin	Pro-growth; stimulates cells in vitro; higher levels ↑ risk, especially in older males	↑
IGF-1	Shares hybrid insulin/IGF-1 receptors; less diet-sensitive; serum and Mendelian randomisation both ↑ risk	↑
Intrauterine oestrogen	Slight excess → enlargement; very high (young animals) → small sclerotic prostate; may imprint the prostate. Higher oestrogen in Black mothers → proposed in-utero contribution	↑ (hypothesis)

- **Trap:** adult serum testosterone does not predict risk; only **genetically determined free testosterone** and functional markers do, and modestly.

Concept — Many "assorted" associations are artefacts of who gets a PSA test (vasectomy, prostatitis, inflammation, STIs); those that survive point to **early-life hormonal programming** — puberty timing, in-utero oestrogen, genetically higher free testosterone — acting decades before diagnosis.

PRIMARY PREVENTION

Drugs

- High prevalence and morbidity make prostate cancer a prime chemoprevention target, yet **no agent is approved by the United States Food and Drug Administration (FDA)** despite several large, often positive, randomised trials.

- **The bar has moved from preventing total to preventing aggressive cancer**; no agent has met it in phase III. Prostate cancer death is rare in healthy men, making benefit hard to show.
- Proposal: target high-risk cohorts and mechanism-matched patients — but predictive biomarkers are lacking, or so clear that withholding the drug is unethical (a statin in hypercholesterolaemia).
- Effort has therefore shifted to **secondary prevention** (delaying progression), largely beyond this chapter.

5 α -Reductase Inhibitors

- 5 α -Reductase inhibitors (5ARIs) — **finasteride, dutasteride** — block testosterone → **dihydrotestosterone (DHT)**; in enlarged prostates they **shrink volume ~25% and lower PSA ~50%**.
- Rationale: androgen dependence; androgen–risk links; **46,XY infants with 5 α -reductase deficiency have only rudimentary prostates**.
- The best-studied and most controversial preventive drugs.

PCPT VS REDUCE — 5ARI CHEMOPREVENTION TRIALS

	Prostate Cancer Prevention Trial (PCPT)	REduction by DUTasteride of prostate Cancer Events (REDUCE)
Agent	Finasteride 5 mg daily vs placebo	Dutasteride 0.5 mg daily vs placebo
Population	Average risk; PSA $\leq$3.0 ng/mL ; 18,882 men	High risk : raised PSA (2.5–10 ng/mL) and a prior negative biopsy
Follow-up / biopsy	7 years; annual PSA and rectal examination; end-of-study biopsy for all	4 years; protocol biopsies at 2 and 4 years , plus for-cause
Primary end point	Prevalence over 7 years	Cancers diagnosed over 4 years
Overall cancer	↓ 24.8% (18.4% vs 24.4%)	↓ 22.8% (19.9% vs 24.9%)
High-grade cancer	Gleason 7–10 ↑: 6.4% vs 5.1%	Gleason 8–10 ↑: 0.9% vs 0.6% ; years 3–4: 12 vs 1
Mortality	18.4-year median follow-up: non-significant 25% — lower prostate cancer death	
Regulatory outcome	Not approved for prevention; FDA added a high-grade cancer warning to 5ARI labels	

- Proposed artefacts for the high-grade excess — **finasteride makes PSA a better marker** (biopsy selection) and **shrinks the gland** (easier detection) — were **not accepted by the FDA**.
- **Trap**: two differently designed trials gave the same answer — about a quarter fewer cancers, but slightly more high-grade ones.

Observational Data

- Meta-analyses of real-world use confirm **lower diagnosis, higher Gleason 8–10**, but **no association with prostate cancer mortality** (>3 million men); null also when excluding baseline cancers and in PSA-adjusted analyses.
- **Conclusion: 5ARIs reduce diagnosis, with no or minimal effect on prostate cancer death.**

Aspirin/Nonsteroidal Anti-inflammatory Drugs

- Rationale: inflammation. **Aspirin**: small reduction in risk, **no effect on mortality**; **non-steroidal anti-inflammatory drugs (NSAIDs)**: similar modest reduction.
- Screening bias (aspirin users are health-seekers) may **underestimate** benefit; within **REDUCE**, where all were biopsied, the reduction was slightly stronger.
- Still modest — **not used routinely for prevention**.

Statins

- Most prescribed drugs in the West. **Cholesterol is the precursor of testosterone** and contributes to inflammation.
- Observational: **lower advanced/fatal cancer** and better outcomes; **not replicated in trials** (short follow-up); **no randomised trial with a prostate cancer prevention end point**. Remain very promising.

Metformin

- Lowers insulin; rationale from the insulin link. Observational data largely null — **no association among diabetics**.

CHEMOPREVENTIVE DRUGS — COMPARISON

Agent	Mechanism	Evidence	Total cancer	Aggressive / fatal	Status
5ARIs	Block testosterone → DHT	Phase III (PCPT, REDUCE) + observational	↓ ~25%	↑ high-grade detection; mortality null	Not approved; warning label
Aspirin / NSAIDs	Anti-inflammatory	Observational; REDUCE secondary analysis	Small ↓	Mortality null	Not routine
Statins	↓ cholesterol (testosterone precursor), anti-inflammatory	Observational only	—	↓ advanced/fatal	Promising; untested in RCT
Metformin	↓ insulin	Observational	Null	—	No evidence

Concept — *Every agent to date reduces the diagnosis of indolent cancer without proven effect on lethal cancer; once the bar became **aggressive disease**, the total-incidence wins of PCPT and REDUCE no longer counted, and the one class that targets advanced disease (statins) has never had its phase III trial.*

KEY POINTS: PRIMARY PREVENTION

- No FDA-approved agent prevents prostate cancer despite large, often positive trials.
- The FDA rejected artefact explanations for the high-grade excess with 5ARIs and modified their labels.
- Long-term PCPT follow-up (median 18.4 years): non-significant 25% lower prostate cancer death with finasteride.
- Statins: observational benefit not yet replicated in trials.

SECONDARY PREVENTION

- High prevalence and long natural history shift interest from **primary prevention (avoiding diagnosis)** to **secondary prevention (avoiding progression)**. Drugs are covered elsewhere; natural agents and lifestyle here.
- **Prostate Cancer Lifestyle Trial**: 93 men on active surveillance (Gleason <7, PSA 4–10 ng/mL); 1-year intervention — low-fat plant-based diet, exercise, stress management, group support — vs usual care; at 2 years, **27% of controls vs 5% of the intervention arm** had needed conventional treatment ($P < 0.05$).
- Hard end points (metastasis, prostate cancer death) come mainly from observational studies.

Physical Activity

- Cohort meta-analysis: higher recreational activity → **~31% lower prostate cancer-specific mortality**.
- **Reverse causation** (progressing cancer limits activity) addressed by a **4–6-year lag**; results persisted, strongest for **vigorous** activity.

- Also protects against **cardiovascular disease, which kills more men with non-metastatic prostate cancer than the cancer does.**
- **2018 United States Physical Activity Guidelines: ≥ 150 min/week moderate or ≥ 75 min/week vigorous aerobic activity;** they concluded regular activity is associated with lower prostate cancer mortality.

Diet

- Few studies. Only factor graded at least suggestive: **whole milk/high-fat dairy (≥ 4 servings/week vs 0–3/month) → higher recurrence and mortality.**
- Inconsistent: red meat and eggs ↑; fish, poultry, tomatoes and cruciferous vegetables ↓ — healthier foods tracking lower risk.
- **Physicians' Health Study** dietary patterns:

WESTERN VS PRUDENT PATTERN AFTER DIAGNOSIS (PHYSICIANS' HEALTH STUDY)

Western (processed and red meat, high-fat dairy, refined grains)	Prudent (vegetables, fruit, fish, legumes, whole grains)
Highest vs lowest quartile: HR 2.53 prostate cancer-specific mortality; HR 1.67 all-cause	36% lower all-cause mortality; prostate cancer-specific mortality lower but not significant

- A randomised trial of a **very-low-carbohydrate diet** (insulin-lowering) in biochemical recurrence **slowed PSA doubling time.**

Tobacco

- Smoking after diagnosis → **higher recurrence/progression and prostate cancer-specific mortality,** consistently.
- Caveat: post-diagnosis smokers smoked for decades before, so the timing of harm is unclear, and benefit from quitting **after** diagnosis is unproven.
- Localised disease: **quit ≥ 10 years → recurrence risk as never-smokers; quit < 10 years → increased.**

Alcohol

- Very limited data. Organ-confined ($\leq T2$) disease: **≥ 2 drinks/day → higher prostate cancer mortality, driven by liquor.**
- HPFS (non-metastatic): **beer ↑; moderate wine (3–7 servings/week), mainly red, protective.** Red wine → lower all-cause mortality in both studies.
- Too limited for a recommendation; **beverage type and drinking pattern** (daily moderate vs binge) likely matter; wine polyphenols may be anti-inflammatory.

Anti-inflammatory

- Given inflammation's proposed role, several anti-inflammatory approaches have been tested against progression.

Pomi-T

- Polyphenol-rich whole-food capsule: **pomegranate, green tea, curcumin (turmeric), broccoli.** 6-month placebo-controlled trial in 203 men on active surveillance/watchful waiting or with biochemical recurrence: **significantly smaller PSA rise in both groups;** well tolerated (slight, non-significant bloating, flatulence, loose stools).

Pomegranate

- Single-arm slowing of PSA rise not confirmed by the placebo-controlled trial → **no strong evidence,** despite wide use in biochemical recurrence.

Phytoestrogen

- See Phytoestrogens under Supplements, Vitamins and Minerals.

Soy

- Limited data; the randomised trial of soy protein showed **no effect on recurrence** in high-risk disease.

SECONDARY PREVENTION — INTERVENTIONS AND VERDICT

Intervention	Evidence	Outcome	Verdict
Comprehensive lifestyle	Randomised (Prostate Cancer Lifestyle Trial)	Fewer men needing treatment on surveillance	Supportive
Physical activity	Cohorts, lag-adjusted; national guidelines	↓ prostate cancer mortality	Recommend
Western vs Prudent diet	Cohort	Western ↑ prostate cancer and all-cause death	Prudent pattern
Whole milk/high-fat dairy	Cohorts	↑ recurrence and mortality	Limit
Low-carbohydrate diet	Randomised	Slower PSA doubling	Promising
Smoking	Cohorts	↑ recurrence and mortality; quit ≥10 years neutralises	Stop
Alcohol	Cohorts	Liquor/beer ↑; red wine ↓	No recommendation
Pomi-T	Randomised	Smaller PSA rise	Promising
Pomegranate; soy	Randomised	Null	No benefit

Concept — After diagnosis the same lifestyle axis recurs — activity, prudent diet, avoiding high-fat dairy and smoking — and much of its benefit is in **competing (cardiovascular) mortality** as well as prostate cancer death; single natural agents again fail placebo-controlled testing, Pomi-T excepted.

SUMMARY

- Very common, especially subclinically (autopsy) → **anything that increases screening increases diagnosis**, hindering identification of true risk factors.
- **Age, family history and race** are unequivocal; family history acts through genetics and shared lifestyle.
- Increasingly strong evidence for **obesity and smoking**, especially for lethal disease.
- Diet probably matters, but the ideal anti-prostate cancer diet remains elusive.
- **No drug is approved or widely used for prevention.**